## Vertex Formula

**1** The $x$-coordinate of the vertex of the parabola occurs at $x = \frac{-b}{2a}$. The $y$-coordinate of the vertex is found by substituting $x = \frac{-b}{2a}$ into the parabola's equation and evaluating the function at this value of $x$.

**2** Axis of symmetry

**3**

a $x = 0, 6$

b $(3, 9)$

**4**

a Concave up

b $x = 9$

c $y = -71$

**5**

a Concave up

b $x = 1, 5$

c $y = 5$

d $x = 3$

e $(3, -4)$

**6**

a Concave down

b $x = -4, 2$

c $y = 8$

d $x = -1$

e $(-1, 9)$

**7**

a Concave down

b $x = -4, -2$

c $y = -8$

d $x = -3$

e $(-3, 1)$

**8**

a $x = -2$

b $-1$

c $(-2, -1)$

d $(-2, -2)$

e

i Yes

ii No

iii Yes

**9**

a $x = 14, 6$

b $x = 10$

c Concave down

d $y = 16$

**10** $(1, -12)$

**11** $y = 0$

**12**

a $-0.087$

b $-1.24$

**13** $x = \dfrac{1}{4}$

**14**

a Vertex of $P_1 = (-2, 2)$
Vertex of $P_2 = (2, 2)$

b $4$

**15**

a $x = -3$

b $-5$

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## Applications

**16**

a $s = 12$

b $337$

**17**

a $t = 3$

b $122$ units